

Introdução às Redes Neurais

(continuação)

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Insper

$$Y \sim \text{Bernoulli}(\theta) \qquad P(Y = 1) = \theta \qquad P(Y = 0) = 1 - \theta$$

$$P(Y = y) = \theta^y (1 - \theta)^{1-y} \qquad y = 0, 1$$

Função de verossimilhança para uma amostra aleatória $Y_1, \dots, Y_n$:

$$\mathcal{L} = P_{\theta}(Y_1 = y_1, \dots, Y_n = y_n) = \prod_{i=1}^n P_{\theta}(Y_i = y_i)$$

$$\mathcal{L} = \theta^{\sum_{i=1}^n y_i} (1 - \theta)^{n - \sum_{i=1}^n y_i}$$

Log-verossimilhança:

$$L = \ln \mathcal{L} = \left(\sum_{i=1}^n y_i \right) \ln \theta + \left(n - \sum_{i=1}^n y_i \right) \ln(1 - \theta)$$

Em Estatística, procuramos o valor do parâmetro θ que maximiza a função de verossimilhança.

Este valor será nossa estimativa de máxima verossimilhança.

Maximizar $\mathcal{L}$ é equivalente a maximizar L , porque o logaritmo é uma função monótona crescente.

Impondo que $dL/d\theta = 0$, é fácil mostrar que

$$\hat{\theta}_{\text{MV}} = \frac{1}{n} \sum_{i=1}^n y_i.$$

$$y_1 = y \qquad y_2 = 1 - y_1$$

$$y = 1 \Rightarrow (y_1, y_2) = (1, 0)$$

$$y = 0 \Rightarrow (y_1, y_2) = (0, 1)$$

$$p_1 = \theta \qquad p_2 = 1 - \theta$$

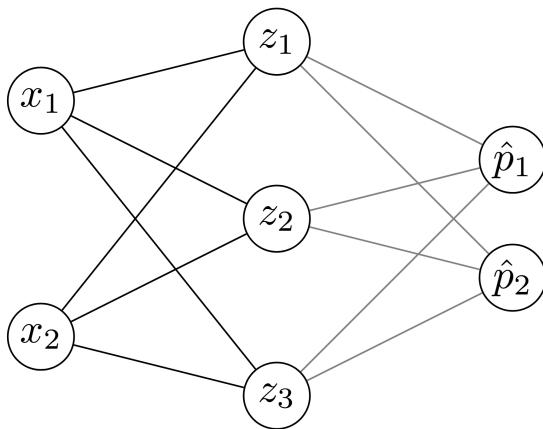
$$\begin{aligned}\mathcal{L} &= P_{(p_1, p_2)}(Y_1 = (y_{11}, y_{12}), \dots, Y_n = (y_{n1}, y_{n2})) \\ &= \prod_{i=1}^n P_{(p_1, p_2)}(Y_i = (y_{i1}, y_{i2})) \\ &= p_1^{\sum_{i=1}^n y_{i1}} \times p_2^{\sum_{i=1}^n y_{i2}}\end{aligned}$$

$$L = \ln \mathcal{L} = \sum_{i=1}^n (y_{i1} \ln p_1 + y_{i2} \ln p_2) = \sum_{i=1}^n \sum_{j=1}^2 y_{ij} \ln p_j$$

Em Machine Learning, esta função de log-verossimilhança escrita com *one-hot-encoding* e multiplicada por -1 é denominada por *entropia cruzada* (*cross-entropy*).

Portanto, se em Estatística o objetivo é maximizar a função de log-verossimilhança, em Machine Learning o objetivo será minimizar a entropia cruzada.

$$\text{Cross-entropy} = -L = - \sum_{i=1}^n \sum_{j=1}^2 y_{ij} \ln p_j$$



$$u_1 = w_1x_1 + w_2x_2 + b_1$$

$$u_2 = w_3x_1 + w_4x_2 + b_2$$

$$u_3 = w_5x_1 + w_6x_2 + b_3$$

$$z_1 = \sigma(u_1)$$

$$z_2 = \sigma(u_2)$$

$$z_3 = \sigma(u_3)$$

$$v_1 = w_7z_1 + w_8z_2 + w_9z_3 + b_4$$

$$v_2 = w_{10}z_1 + w_{11}z_2 + w_{12}z_3 + b_5$$

$$\hat{p}_1 = \frac{e^{v_1}}{e^{v_1} + e^{v_2}}$$

$$\hat{p}_2 = \frac{e^{v_2}}{e^{v_1} + e^{v_2}}$$

$$L = -(y_1 \ln \hat{p}_1 + y_2 \ln \hat{p}_2)$$

A entropia cruzada L depende do peso w_1 por *quatro caminhos*:

$$A_1 : w_1 \rightarrow u_1 \rightarrow z_1 \rightarrow v_1 \rightarrow \hat{p}_1 \rightarrow L$$

$$A_2 : w_1 \rightarrow u_1 \rightarrow z_1 \rightarrow v_2 \rightarrow \hat{p}_1 \rightarrow L$$

$$A_3 : w_1 \rightarrow u_1 \rightarrow z_1 \rightarrow v_1 \rightarrow \hat{p}_2 \rightarrow L$$

$$A_4 : w_1 \rightarrow u_1 \rightarrow z_1 \rightarrow v_2 \rightarrow \hat{p}_2 \rightarrow L$$

A derivada parcial desejada será a soma destes quatro produtos (cada um deles obtidos via regra da cadeia):

$$\frac{\partial L}{\partial w_1} = A_1 + A_2 + A_3 + A_4$$

$$\begin{aligned}
 \frac{\partial L}{\partial w_1} &= \frac{\partial L}{\partial \hat{p}_1} \frac{\partial \hat{p}_1}{\partial w_1} + \frac{\partial L}{\partial \hat{p}_2} \frac{\partial \hat{p}_2}{\partial w_1} \\
 &= \frac{\partial L}{\partial \hat{p}_1} \left(\frac{\partial \hat{p}_1}{\partial v_1} \frac{\partial v_1}{\partial w_1} + \frac{\partial \hat{p}_1}{\partial v_2} \frac{\partial v_2}{\partial w_1} \right) + \frac{\partial L}{\partial \hat{p}_2} \left(\frac{\partial \hat{p}_2}{\partial v_1} \frac{\partial v_1}{\partial w_1} + \frac{\partial \hat{p}_2}{\partial v_2} \frac{\partial v_2}{\partial w_1} \right) \\
 &= \frac{\partial L}{\partial \hat{p}_1} \frac{\partial \hat{p}_1}{\partial v_1} \frac{\partial v_1}{\partial w_1} + \frac{\partial L}{\partial \hat{p}_1} \frac{\partial \hat{p}_1}{\partial v_2} \frac{\partial v_2}{\partial w_1} + \frac{\partial L}{\partial \hat{p}_2} \frac{\partial \hat{p}_2}{\partial v_1} \frac{\partial v_1}{\partial w_1} + \frac{\partial L}{\partial \hat{p}_2} \frac{\partial \hat{p}_2}{\partial v_2} \frac{\partial v_2}{\partial w_1} \\
 &= \frac{\partial L}{\partial \hat{p}_1} \frac{\partial \hat{p}_1}{\partial v_1} \frac{\partial v_1}{\partial z_1} \frac{\partial z_1}{\partial u_1} \frac{\partial u_1}{\partial w_1} + \frac{\partial L}{\partial \hat{p}_1} \frac{\partial \hat{p}_1}{\partial v_2} \frac{\partial v_2}{\partial z_1} \frac{\partial z_1}{\partial u_1} \frac{\partial u_1}{\partial w_1} \\
 &\quad + \frac{\partial L}{\partial \hat{p}_2} \frac{\partial \hat{p}_2}{\partial v_1} \frac{\partial v_1}{\partial z_1} \frac{\partial z_1}{\partial u_1} \frac{\partial u_1}{\partial w_1} + \frac{\partial L}{\partial \hat{p}_2} \frac{\partial \hat{p}_2}{\partial v_2} \frac{\partial v_2}{\partial z_1} \frac{\partial z_1}{\partial u_1} \frac{\partial u_1}{\partial w_1}
 \end{aligned}$$

Derivadas parciais elementares envolvidas nos cálculos de A_1 , A_2 , A_3 e A_4 :

$$\partial u_1 / \partial w_1 = x_1$$

$$\partial z_1 / \partial u_1 = \sigma'(u_1)$$

$$\partial v_1 / \partial z_1 = w_7$$

$$\partial v_2 / \partial z_1 = w_{10}$$

$$\partial \hat{p}_1 / \partial v_1 = \hat{p}_1(1 - \hat{p}_1)$$

$$\partial \hat{p}_1 / \partial v_2 = -\hat{p}_1 \hat{p}_2$$

$$\partial \hat{p}_2 / \partial v_1 = -\hat{p}_1 \hat{p}_2$$

$$\partial \hat{p}_2 / \partial v_2 = \hat{p}_2(1 - \hat{p}_2)$$

$$\partial L / \partial \hat{p}_1 = -y_1 / \hat{p}_1$$

$$\partial L / \partial \hat{p}_2 = -y_2 / \hat{p}_2$$

$$\frac{\partial L}{\partial w_1} = ((\hat{p}_1 - y_1) w_7 + (\hat{p}_2 - y_2) w_{10}) \sigma'(u_1) x_1$$

$$\frac{\partial L}{\partial w_2} = ((\hat{p}_1 - y_1) w_7 + (\hat{p}_2 - y_2) w_{10}) \sigma'(u_1) x_2$$

$$\frac{\partial L}{\partial b_1} = ((\hat{p}_1 - y_1) w_7 + (\hat{p}_2 - y_2) w_{10}) \sigma'(u_1)$$

$$\frac{\partial L}{\partial w_3} = ((\hat{p}_1 - y_1) w_8 + (\hat{p}_2 - y_2) w_{11}) \sigma'(u_2) x_1$$

$$\frac{\partial L}{\partial w_4} = ((\hat{p}_1 - y_1) w_8 + (\hat{p}_2 - y_2) w_{11}) \sigma'(u_2) x_2$$

$$\frac{\partial L}{\partial b_2} = ((\hat{p}_1 - y_1) w_8 + (\hat{p}_2 - y_2) w_{11}) \sigma'(u_2)$$

$$\frac{\partial L}{\partial w_5} = ((\hat{p}_1 - y_1) w_9 + (\hat{p}_2 - y_2) w_{12}) \sigma'(u_3) x_1$$

$$\frac{\partial L}{\partial w_6} = ((\hat{p}_1 - y_1) w_9 + (\hat{p}_2 - y_2) w_{12}) \sigma'(u_3) x_2$$

$$\frac{\partial L}{\partial b_3} = ((\hat{p}_1 - y_1) w_9 + (\hat{p}_2 - y_2) w_{12}) \sigma'(u_3)$$

$$\frac{\partial L}{\partial w_7} = (\hat{p}_1 - y_1) z_1$$

$$\frac{\partial L}{\partial w_8} = (\hat{p}_1 - y_1) z_2$$

$$\frac{\partial L}{\partial w_9} = (\hat{p}_1 - y_1) z_3$$

$$\frac{\partial L}{\partial b_4} = \hat{p}_1 - y_1$$

$$\begin{aligned}\frac{\partial L}{\partial w_{10}} &= (\hat{p}_2 - y_2) z_1 \\ \frac{\partial L}{\partial w_{11}} &= (\hat{p}_2 - y_2) z_2 \\ \frac{\partial L}{\partial w_{12}} &= (\hat{p}_2 - y_2) z_3 \\ \frac{\partial L}{\partial b_5} &= \hat{p}_2 - y_2\end{aligned}$$